

Guru Naveen Reddy Siddu

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ASIC Design Verification Engineer — Verilog — SystemVerilog — UVM — SVA

Professional Summary

- Design Verification Engineer who took a **PCIe Gen-2 SerDes** block from RTL to **100% functional coverage and verification sign-off** using a self-built UVM environment.
- Builds reusable UVM verification environments end-to-end – sequencers, drivers, monitors, scoreboards, agents, and coverage models – for protocol-level IP verification (PCIe, APB, UART).
- Skilled in constrained-random verification, assertion-based verification (SVA), coverage-driven verification, regression execution, waveform debugging, and root-cause analysis using QuestaSim.
- Strong grounding in Verilog RTL design, Linux, Git, and Python scripting, with hands-on protocol knowledge spanning LTSSM, 8b/10b encoding, and AMBA APB.

Technical Skills

HDL / HVL: Verilog, SystemVerilog, SystemVerilog Assertions (SVA), Python.

Verification Methodologies: UVM, Functional Verification, Constrained Random Verification, Assertion Based Verification (ABV), Coverage Driven Verification (CDV), Functional Coverage, Code Coverage, Coverage Closure, Regression Testing.

UVM Concepts: Sequence Item, Sequence, Sequencer, Driver, Monitor, Agent, Environment, Scoreboard, Test, Virtual Sequence, Factory Override, Config DB, Analysis Port, TLM Communication, Virtual Interface, UVM Phases.

Protocols / IPs: PCIe Gen-2 (LTSSM, 8b/10b Encoding, SerDes), APB, UART.

EDA Tools: QuestaSim, ModelSim.

Version Control: Git

Operating System: Linux

Projects

PCIe Gen-2 SerDes – Design & Verification *Verilog — SystemVerilog — UVM — QuestaSim*

Designed and verified the PCIe Gen-2 Physical Layer SerDes – transmitter and receiver sub-blocks – with a reusable UVM environment built to validate protocol compliance and data integrity.

- Designed **Scrambler, 8b/10b Encoder, PISO, SIPO, 8b/10b Decoder, and Descrambler** sub-blocks in Verilog.
- Built a reusable **UVM environment** (sequencer, driver, monitor, scoreboard, agent, env, tests) from scratch.
- Wrote **SVA** and functional coverage models to verify **LTSSM, 8b/10b encoding/decoding, TLPs, DLLPs**, and Configuration Space (Type-0/Type-1) behavior.
- Achieved **100% functional coverage** and closed verification sign-off through regression execution and waveform-based root-cause debugging in QuestaSim.

Verified the APB FSM (Idle → Setup → Access) across single- and multi-slave communication, with focus on **PREADY**-driven wait-state handling.

- Architected a class-based UVM environment from a self-authored verification plan / feature extraction sheet.
- Verified write/read operations with and without wait states, same-slave and cross-slave communication, and **20 back-to-back** read/write transactions.
- Covered random, constant, increment, and decrement address/data patterns to stress-test protocol edge cases.
- Closed **functional and code coverage** for sign-off; wrote SVA for reset, read/write, and state-transition checks.

UART Receiver with FIFO – RTL Design & Verification *Verilog — SystemVerilog — QuestaSim*

Designed a UART Receiver with an integrated synchronous FIFO for reliable serial reception and buffering, verified via a SystemVerilog testbench.

- Designed a **5-state FSM** (Idle, Start, Data, Parity, Stop) in Verilog with parity checking and stop-bit validation.
- Built a **parameterized synchronous FIFO** with pointer logic, occupancy counter, and full/empty flags for configurable depth and width.
- Integrated UART-to-FIFO data path and validated frame reception, buffering, and reset behavior across multiple transmission scenarios.
- Confirmed correct data transfer through simulation and waveform debugging in QuestaSim.

Certifications & Training

- **RTL Design & Verification Training Certificate** – VLSI FIRST Technology, Bengaluru
- Hands-on program in ASIC RTL Design and Functional Verification – Verilog, SystemVerilog, UVM, SVA – applied directly to the PCIe Gen-2, APB, and UART projects above.

Education

B.Tech in Electronics & Communication Engineering
Madanapalle Institute of Technology and Science, Andhra Pradesh

2021 – 2024
CGPA: 8.4 / 10.0